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Low-density $Fe_{40}Mn_{19}Ni_{15}Al_{15}Si_{10}C_1$ high entropy steel processed by mechanical alloying and spark plasma sintering: Phase evolution, microstructure and mechanical properties

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ABSTRACT

The present study deals with the synthesis of a novel $Fe_{40}Mn_{19}Ni_{15}Al_{15}Si_{10}C_1$ low-density high entropy steel (HES) processed through mechanical alloying (MA) and spark plasma sintering (SPS). The elemental powder was milled for 35 h, leading to the formation of multiphase structures i.e., BCC phase (a = 0.286 nm; cI2), γ-brass type (a = 0.872 nm; cI52) and partially ordered B2-type (a = 0.290 nm; cP2) phases with the trace amount of Si. The milled powder was found to be thermally stable up to ~500 °C, however, the formation of Fe_5Si_3-type phase was evident at ~520 °C. The spark plasma sintered (SPSed) sample was able to retain the BCC phase (a = 0.286 nm), γ-brass type (a = 0.872 nm), B2 (a = 0.289 nm) along with the formation of Fe_5Si_3 type silicide phase (a = b = 0.667 nm, c = 0.468 nm; hP16). These SPSed HES sample was found to have low density (~6.49 ± 0.3 g cm^{-3}), high microhardness (~7.8 ± 0.3 GPa) and good compressive strength (~2046 ± 160 MPa) with an appreciable ductility of ~19%. The enhanced mechanical properties of the SPSed sample can be attributed to dual-phase microstructure i.e., BCC and B2 along with finer nano-sized silicide precipitates leading to dominant strengthening mechanisms (i.e., grain-boundary and dislocation strengthening). The experimental findings on phase evolution, thermal stability and mechanical properties of non-equiatomic HES were correlated with the various thermodynamic parameters, CALPHAD modelling and strengthening mechanisms. The significance of the present study is to design a high entropy steel containing the dual phase microstructures i.e., BCC as major phase and B2 as secondary phase, for giving rise to high mechanical strength and high temperature stability.

1. Introduction

The thrust for development of advanced materials having a synergy between strength and ductility has been evolving in the modern era of industrialization. The urge for high strength advanced materials led to the emergence of high entropy alloys (HEAs). The HEAs are usually disordered single-phase solid solution consisting of five or more major alloying elements in equiatomic or near-equiatomic compositions [1]. HEAs have allured the materials industries across the world due to their excellent mechanical and functional properties, and attracted attention from the research community [2]. These HEAs can be sub-categorized into various classes based on the types of alloying elements, microstructure and/or phases formed i.e., eutectic [3], non-equiatomic [4], precipitation strengthened [5], interstitial [6] HEAs, TRIP and TWIP HEAs [7,8], high entropy intermetallics [9], high entropy nanoparticles [10], and high entropy steels [11].

The HEAs are mainly prepared through liquid metallurgy route i.e., vacuum arc melting or vacuum induction melting. However, this route has some limitations owing to porosity, segregation, difference in melting point and density among thealloying elements. Powder metallurgy route (i.e., mechanical alloying (MA) followed by sintering.) have overcome these problems and also easily produced the nanocrystalline materials with homogenous solid solution [12]. In addition, this technique can expand the solid solubility even in the immiscible alloys [13].

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Spark plasma sintering (SPS) has become a popular technique for consolidation of mechanically alloyed powders. The high heating rate during SPS can be attributed to the generation of high-power current (pulsed). The samples consolidated by SPS can effectively retain the nanocrystalline features [14]. A few researchers revealed the nanocrystalline nature of FCC phase along with Cr_7C_3 in CrFeCoNi and CrMnFeCoNi HEAs processed by mechanical alloying (MA) and SPS [15]. These HEAs based composites' mechanical properties were almost similar to those of a few advanced composites prepared by powder metallurgical processing e.g., Cu–SiC, Al_2O_3-WC-CO [16,17].

The structural material used for high temperature applications ought to have phase stability up to higher temperatures. Therefore, it is of paramount importance to establish the thermal stability of these HEAs to ascertain two aspects i.e., phase transformation and the retention of nanosized structure. The CrFeCoNi HEA has revealed the high resistance to grain growth even at long duration of heat treatment at 900 °C for 600 h [15]. Shivam et al. [18] synthesized AlCoCrFeMnNi HEA by mechano-chemical process and illustrated that the powder sample was thermally stable up to 500 °C and afterwards encountered diffusional transformation to form minor fraction of intermetallics along with the parent phase at room temperature.

The use of carbon as an important interstitial alloying element has strong effect on the phase constitution and mechanical properties on steels, Ni-based and Co-based superalloys [19]. The addition of carbon in $Fe_{40.4}Ni_{11.3}Mn_{34.8}Al_{7.5}Cr_6$ [20], $Fe_{40}Mn_{40}Co_{10}Cr_{10}$ [21], FeCoNiAl [22], $Ni_{1.5}CoCuFeCr_{0.5\text{-}x}V_x$ (x = 0.25, 0.5 mol) [23], FeCoNiCu [24], TiFeCoNiAl [25] and $Cr_{20}Mn_{10}Fe_{35}Ni_{35}$ [26] HEAs showed excellent mechanical properties at room and high temperatures. Raabe et al. [27] studied the FeMnAl–C and FeMnAlSi–C high entropy steel having FCC structure and exhibiting good tensile ductility with nominal strength [27]. They explained the formation of different phases by varying the C and Mn content in the non equiatomic FeMnAlSi–C high entropy steel. The effect of C and Mn in the phase formation of FeMnAlSi–C high entropy steel was shown efrom multiphase to single phase FCC solid solution. Raabe and co-workers illustrated the formation of FCC structure strengthened by addition of Al and Si due to solid solution strengthening [27].

In the view of above, the present investigation envisages the preparation of a non-equiatomic low-density $Fe_{40}Mn_{19}Ni_{15}Al_{15}Si_{10}C_1$ high entropy steel by MA and SPS possessing a multi-phase microstructure consisting of mainly BCC and B2 phases that may have greater strength with appreciable amount of ductility. The alloying behavior, thermal stability, microstructural and chemical analysis, mechanical properties (i.e., microhardness and compressive strength) during MA and SPS have been studied systematically. Further, the mechanical properties of these HES were understood with the help of various strengthening mechanisms.

2. Experimental details

The starting elemental powder of Fe, Ni, Mn, Al, Si and C (99.2% pure) was taken in appropriate proportion of $Fe_{40}Mn_{19}Ni_{15}Al_{15}Si_{10}C_1$ (at. %) (Fe-46.00%, Mn-21.50%, Ni-18.13%, Al-8.34%, Si-5.78%, C-0.25%; wt. %) and was milled in planetary ball mill (Model no. PM 400/2, Retsch) at the speed of 200 rpm. The ball to powder ratio was taken as 10:1. The milling was carried out in the tungsten carbide (WC) vials and with WC balls. To avoid oxidation, toluene was taken as a process control reagent during milling. To understand the phase evolution during milling, 2–3 g of milled powder was taken after every 5 h of milling. The milling was stopped intermittently for 20 min after every 1 h for avoiding any overheating. The 35 h milled powder was consolidated by spark plasma sintering (Model no. Dr. Sinter SCM 1050, Japan) at a temperature of 900 °C, pressure of 50 MPa, and heating rate of 100 °C. min^{-1} with a holding time of 15 min. The diameter and thickness of the spark plasma sintered sample (SPSed) was ~30 mm and ~9 mm, respectively. The pellet was cleaned with the help of polisher to remove the graphite layer. The wire EDM machine (Medha enterprises, model no. EXPRESSCUT EX-2530) was used to cut the pellet into many pieces of dimensions (diameter = 4 mm and height = 8 mm) for further characterization and evaluation of mechanical properties. The consolidated pellet was found to have a relative density of ~99%, which was calcualted using the following [28]:

$$\rho_{RD} = \frac{\rho_{exp}}{\rho_{th}} \times 100 \quad (1)$$

where, ρ_{RD} is relative density (%), ρ_{th} is therotecial density computed by the rule of mixture (6.55 g/cc), and ρ_{exp} is experimental density calculated by the Archmides principle [29]. The experiment was repeated six times to calculate the average experimental density (6.49 ± 0.3 g/cc).

The differential scanning calorimeter (DSC) (Netzsch, DSC404 Pegasus) was used to study the thermal stability of milled powder. The 35 h milled powder was heated up to 1000 °C with 20 K/min of heating rate under the nitrogen (N_2) atomsphere. 35 h milled powder were encapsulated in a quartz tube and back filled with argon atmosphere for annealing it at higher temperatures. For better understanding of the thermal stability of this high entropy steel, 35 h milled sample was annealed for 1 h at diferent temperatures (i.e., 400 °C, 500 °C, 600 °C and 800 °C) followed by water quenching. The alloying behavior during milling and phase formation after the SPS was ascertained by using high resolution X-Ray diffraction (XRD) (EMPYREAN, PANALYTICAL) with a Co-Kα source (operating voltage:40 kV; operating current:40 mA) with the scan rate of 5° min^{-1}. The XRD analysis of the annealed sample was done in the Rigaku X-Ray diffractometer (Cu- Kα source) operating at voltage of 40 kV and 15 mA of current with the scan rate of 5° min^{-1}. The microstructural feature, chemical analysis and mapping of the milled powder at different milling time was examined using Evo 18 Carl Zeiss scanning electron microscope (SEM), and field emission scanning electron microscopee (FE-SEM) (FEI model no. Nova Nano SEM 450), both equipped with EDS detector. The microstructural study of the SPSed pellet was done with the help of FE-SEM (model no. Nova Nano SEM 450). The elemental analysis of the SPSed sample was performed by the EDS detector attached with SEM. For microstructural features, the SPSed samples were systematically polished and chemically etched with the 5% nitric acid and 95% methanol solution (Nital). For 35 h milled sample, structural analysis was done with the help of transmission electron microscope (TEM) (TECNAI G^2T20, FEI) operating at 200 kV. Electron Probe Micro Analyzer (EPMA) (CAMECA SXFive) was used for BSE imaging, quantitative point analysis and X-ray elemental mapping of the SPSed sample. The SPSed samples were carbon coated with 20 nm layer using LEICA-EM ACE200 instrument. Wavelength-dispersive spectrometry (WDS), LaB6 filament and TAP, PET, LIF, LPET and LTAP crystals were used at a voltage of 15 kV and current of 10 nA for the analyses. The acquisition, data processing and routine calibration were carried out using SxSAB version 6.1 and SX-Results software of CAMECA.

The XRD data of the milled samples, annealed and SPSed pellet were refined using Rietveld refinement method in the general structural analysis system (GSAS-II) [30] to determine the structural (lattice parameter) and microstructural (crystallite size and microstrain) parameters. The background was fitted with the Chebyschev-I background function with 10 coefficients. The peak profiles were fitted from broadening of Lorentzian and Gaussian component of pseudo-Voigt profile function. The phase fraction was determined using histogram scale factor. The average crystallite size and isotropic microstrain was based on uniaxial model [30]. The least-square method was used in order to minimize the difference between the experimental and observed data to achieve the acceptable goodness of fit (GOF) factor [31]. The dislocation density [32] was calculated from the equation,

$$\rho_{dd} = \frac{\left(3\sqrt{2\pi} < \varepsilon^2 >^{\frac{1}{2}}\right)}{tb} \quad (2)$$

where, ρ_{dd} is dislocation density, t is crystallite size in nm, b is Burger's vector $= \frac{\sqrt{3}}{2}$ a for BCC, a = lattice parameter in nm, ε = microstrain. The microstrain [33] was calculated using equation,

$$\beta = 4\varepsilon \tan\theta \quad (3)$$

where, β is FWHM and θ is Bragg angle.

The mechanical properties i.e., microhardness and uniaxial compression test of the SPSed samples were done using the MHT^3 (Make: Anton Paar) instrumented microhardness tester and 100 kN Universal Testing Machine (Make: Instron; model no. 5982) respectively. The microhardness test was performed with the load of 5000 mN. The dwell time of 20 s, approach speed of 80 μm/min, retract speed of 16 μm/min, loading and unloading rate of 10,000 mN/min, respevtively. The Vickers diamond indentator was used and atleast six measurements were done to measure the microhardness. To determine the microhardness and elastic modulus of the SPSed pellets, Oliver-Pharr method was used [34]. The compression test was done at 25 °C with the sample dimension of diameter (4 mm) to height (8 mm) ratio of 1:2 and at crosshead speed of 0.5 mm/min with the strain rate of 10^{-3} s^{-1}. The test was repeated three times to evaluate the average ultimate compressive strength.

3. Results

The melting point of the high entropy steel was calculated with the help of machine learning (ML) approach formulae [35,36] and rule of mixture. The melting point obtained from ML and rule of mixture was found to be 1638 K and 1620 K, respectively. The ML formulae used to calculate the melting point [36], is as follows,

$$T_m = \frac{\sum_{i \neq j} T_{i-j} \times C_i \times C_j}{\sum_{i \neq j} C_i \times C_j} \quad (4)$$

where, T_{ij} = binary liquidus temperature (°C) at the element relative ratio (C_i: Cj), C_i and C_j are the molar percentages of the i and j elements, respectively. The value of the liquidus temperature of all the binary alloys [37] used for computing the melting point of HES composition are given in Table 1. The binary enthalpy of mixing were also calculated using semi-empirical Miedema approach [38] and compiled in Table 1. It was reported that the prediction of the melting point of the HEAs through machine learning approach was more precise and accurate as compared to the rule of mixture [36].

Phase evolution study of the milled samples are studied using XRD at various milling time, and it is shown in Fig. 1(a). The pre-mixed powder of alloying elements at 0 h reveals the diffraction peaks related to the elements i.e., Fe (cI2, JCPDS# 00-006-0696), Mn (cI58, JCPDS# 00-032-0637), Ni (cF4, JCPDS# 00-004-0850), Al (cF4, JCPDS# 00-004-0787), Si (cF8, JCPDS# 00-005-0565) and C (hP4, JCPDS# 00-056-0159). It is evident (from Fig. 1(a)) that the diffraction peak intensity of all the element decreases with the milling time. Low intensity peaks of all the elements have vanished after 5 h of milling. The intensity of Al diminishes after 15 h of milling indicating that Al has the highest alloying rate. After 20 h, intense peaks of Ni and Mn are not evident in the XRD pattern. The alloying of the elements with the milling time can be related to the diffusion of low melting point elements into the matrix of higher melting point elements, as reported previously [39]. The solid solution has been formed after 25 h of milling. The phases formed are BCC (a = 0.286 nm; JCPDS# 00-006-0696) and γ-brass type (a = 0.872 nm; JCPDS# 01-078-2831) structure. The milling has been extended up to 35 h, for further nanostructuring of milled powder and formation of

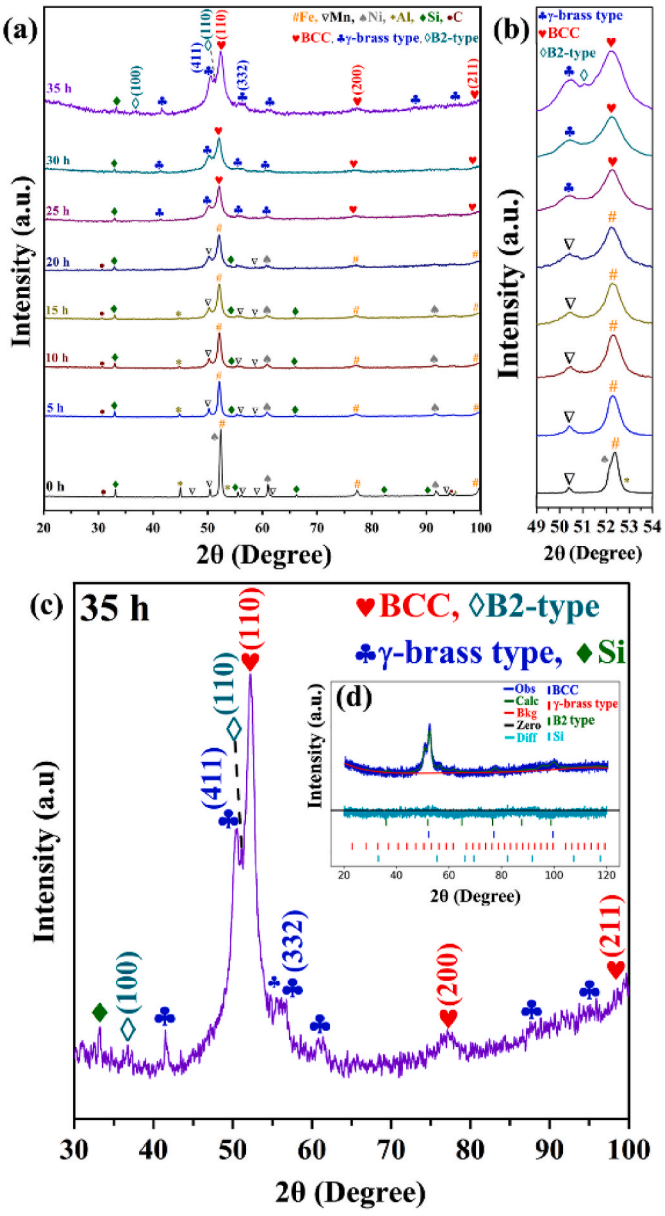


Fig. 1. (a) XRD patterns of the milled samples at various milling time; (b) Enlarged view in the range of 2θ (degree) = 49°-54°; (c) XRD pattern of the 35 h milled sample; (d) Rietveld refinement of the corresponding 35 h milled sample. This shows the alloying behavior of alloy as a function of milling time, and formation of the multi-phase structure (ferrite as a major phase) after 35 h of milling.

Table 1
The liquidus temperature of binary alloys used for computing the melting point of high entropy steel by machine learning approach and binary mixing enthalpy of the constituent elements by semi-empirical Midema approach.

Binary Alloys	Binary Enthalpy (kJ/mol)	Liquidus Temperature (°C)
Fe–Mn	1	1420
Fe–Ni	−4	1440
Fe–Al	−39	1500
Fe–Si	−44	1330
Fe–C	−60	1480
Mn–Ni	−13	1050
Mn–Al	−40	1190
Mn–Si	−44	1283
Mn–C	−38	1231
Ni–Al	−40	1638
Ni–Si	−29	1250
Ni–C	−38	1325
Al–Si	−4	950
Al–C	−13	2100
Si–C	−4	2100

homogenous solid solution. There is a phase change i.e., B2-type ordered and coexisted with the other phases. Minor peak broadening and the decrease in the amount of silicon has occurred as milling extended to 35 h.

It is evident that the trace amount of Si remained even after 35 h of milling. The BCC as a major phase, γ-brass type and B2-type (a = 0.290 nm, JCPDS# 01-083-3994) phases along with trace amount of Si were formed after 35 h of milling. In the 35 h milled powder, no signature of contamination was observed, as discerned through X-Ray energy dispersive spectroscopy. In our previous investigation on the Al matrix and quasicrystalline matrix composites milled for more than 50 h in WC vials and with WC balls, no signature of contamination was detected [40,41]. Fig. 1(b) illustrates the XRD pattern with the milling in the enlarged view of 2θ ranges from 49°-54°. For 35 h milled powder, XRD pattern is illustrated in Fig. 1(c) which shows the formation of multi-phase structure. Fig. 1(d) displays the Rietveld refinement plot of 35 h milled powder of the alloy. The standard peaks of the elements (Fe, Mn, Ni, Al and Si) and the XRD pattern of pre-milled powder in Supplementary Fig. S1 (a & b). Similarly, the simulated diffraction peaks of the three phases (BCC, B2-type and γ-brass type) along with the XRD pattern of 35 h milled powder is shown in Supplementary Fig. S2 (a & b). It is clearly evident from Fig. S1 many diffraction peaks of the alloying elements co-exist at the almost the same d-spacing. Similarly, simulated diffraction peaks for BCC, B2-type and γ-brass type (Fig. S2) discerns co-existence of diffractions peaks at almost same d-spacing.

The crystallite size and lattice strain of the milled samples were calculated using the Rietveld refinement method with the help of GSAS-II (Table 2). The crystallite size was found to vary as a function of milling (Table 2). The crystallite size was found to be tuned in the range of ~13–17 nm and ~10–13 nm for BCC and γ-brass type phase respectively. The crystallite size for Si decreased from 20 nm to 14 nm as milling extended from 25 h to 35 h. The lattice strain increased for all the phases as the milling time progressed. The increase in the lattice strain with milling is due to the atomic size difference between the individual elements, mechanical deformation in the powder and the increase in the fraction of grain boundary. Peak broadening and the decrease in diffraction peaks intensity with the milling are evident from Fig. 1(a). The reason is due to the lattice distortion leading to the induced lattice strain and extra grain refinement. The usual characteristics of the milling process and the combination of both the phenomenon contributes to the peak broadening. There is a slight change in the amount of phase fraction for the BCC and γ-brass type structure. The dislocation density shows the negligible change with the milling time and the values are displayed in Table 2.

Fig. 2(a–d) illustrates the size, shape and morphology of the milled powder at different milling time. Three different contrasts i.e., white, light and dark grey are clearly visible in the micrograph of 35 h milled sample. The size and shape of the powder particles are random and bulky at the early duration of the milling (as seen in Fig. 2(a)). The average particle sizes decreased drastically after 20 h of milling. The powder particles were becoming more homogenous as the milling increased. For 35 h milled samples, the mean particle size is 3–5 μm, and the size and shapes of the particles are different. The full area EDS analysis of the marked area (in Fig. 2(d)) is illustrated in Fig. 2(e), showing the peaks of all the elements in 35 h milled powder. Fig. 2(f–i) displays the histogram of particle size distribution at 10 h, 20 h, 30 h and 35 h, respectively. The particles size was measured using ImageJ software. Fig. 2(j) shows the microstructure of the powder pellet (35 h) and its corresponding mapping and the point EDS. Table 3 lists the elemental composition of 35 h milled powder from different areas i.e., full area, spot 1 and spot 2. The light grey contrasts are rich in Fe and it corresponds to BCC (ferritic phase) structure. The dark grey contrasts are rich in Mn and its stoichiometry are close to the γ-brass type structure. The chemical analysis also confirmed the formation of BCC (Fe-rich) and γ-brass type (Mn-rich) in 35-h milled powder.

The bright-field TEM image and its corresponding selected area diffraction pattern are shown in Fig. 3(a) and (b), respectively. The image shows that the nanocrystalline grains were formed after 35 h milled powder. The formation of ring pattern represents the polycrystalline nature of the alloy. The ring diffraction pattern correspond to the BCC crystal structure. The planes of (110), (200) and (211) of the BCC phase can be identified from the ring patterns.

The DTA curve of 35 h milled powder is displayed as Fig. 4, which reveals various exothermic events at 420 °C (693 K) and 520 °C (793 K). To understand the phase transformation of 35 h milled powder, the annealing was done at different temperatures of 400 °C, 500 °C, 600 °C and 800 °C followed by ex-situ XRD of annealed powders (Fig. 5 (a)). The enlarged view of the annealed sample in the 2θ range of 40°- 46° is shown in Fig. 5(b). The deconvolution of phases around (110) peak of BCC phase for samples annealed 400 °C, 500 °C, 600 °C, and 800 °C are presented in Fig. 5(c), (d), (e), and (f), respectively. The exothermic heating event observed at ~420 °C may be corresponding to the ordering and/or texturing of γ-brass type phase (Figs. 4 & 5(d)). The heating event at ~520 °C corresponds to the formation of Fe_5Si_3-type phase (Figs. 4 & 5(e)). It is clear that 35 h milled powder is thermally stable up to 500 °C, and afterwards the diffraction peaks of the Fe_5Si_3 type intermetallic phase (hP16) is evolved at 600 °C and grows further up to 800 °C. At 800 °C, no phase change has occurred but the amount of the B2-type phase is increased. The phase fraction, crystallite size and lattice parameter of the phases evolved during annealing at different tempeatures were ascertained through rietveld refinement and are mentioned in Table 4.

XRD pattern of SPSed sample is shown in Fig. 6(a) and its deconvulated image in the 2θ range of 51°- 56° is displayed in Fig. 6(b). The BCC (ferrite) phase was observed as the major phase along with partially ordered B2 and the trace amount of γ-brass type phase as well as Fe_5Si_3 type phase. Fig. 6(c) shows the Rietveld refinement of the SPSed sample confirming the presence of BCC, B2 type, γ-brass type and Fe_5Si_3 type silicide. The lattice parameter, crystallite size and the phase fraction of the SPSed sample was ascertained through rietveld refinement and is mentioned in Table 4. The SEM micrographs of the SPSed sample at 900 °C and 50 MPa with different magnification are illustrated in Fig. 7(a and b), exhibiting different contrasts i.e., white, black, light and dark grey, which may correspond to the BCC, B2 type, γ-brass type and Fe_5Si_3 type intermetallic phases, respectively. The SEM-EDS at full-area of SPSed pellet is shown in Fig. 7(c). The presence of all the elements in

Table 2
Variation in phase fraction, lattice strain, crystallite size and dislocation density with milling of non-equiatomic HES.

Parameters	25 h; GoF = 1.08, χ^2 = 1.18			30 h; GoF = 1.14, χ^2 = 1.29			35 h; GoF = 1.10, χ^2 = 1.22			
	BCC	γ-brass type	Si	BCC	γ-brass type	Si	BCC	γ-brass type	B2-type	Si
Lattice Parameter (nm)	a = 0.286	a = 0.872	a = 0.543	a = 0.286	a = 0.872	a = 0.543	a = 0.286	a = 0.872	a = 0.290	a = 0.543
Volume (Å^3)	23.39	698.33	160.10	23.39	698.33	160.10	23.39	698.33	24.39	160.10
Weight fraction	74.4	21.0	4.60	73.0	23.5	3.5	64.0	27.0	6.0	3.0
Crystallite Size (nm)	17	13	20	15	12	13	13	10	13	14
Microstrain (× 10^{-4})	10,849	13,731	10,106	13,195	15,445	13,044	17,013	18,829	15,052	18,546
Dislocation density (m^{-2}) (× 10^{18})	1.94	–	–	2.67	–	–	3.97	–	–	–

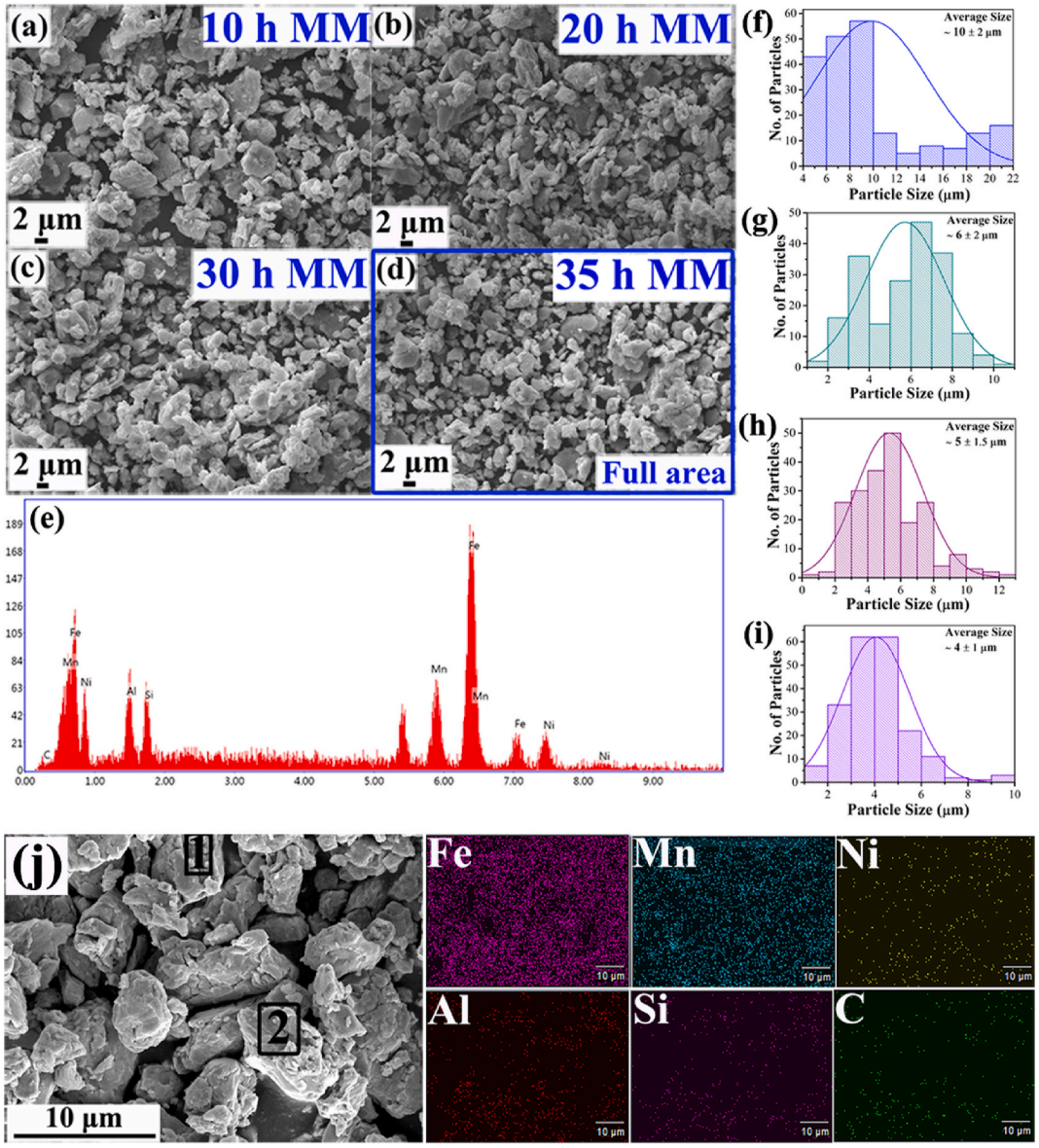


Fig. 2. (a–d) SEM micrographs showing morphology of powder samples milled at different milling time. Fig. 2(d) SEM micrograph showing 35 h milled powder particles used for the full area EDS analysis; (e) EDS spectrum of 35 h milled powder showing presence of different elements in the final composition; (f–i) Histograms showing the particle size distribution of milled samples at 10 h, 20 h, 30 h and 35 h, respectively; (j) SEM micrograph of the 35 h milled powder, marked for point EDS analysis and EDS mapping. The SEM micrograph illustrates the different contrasts i.e., light and dark grey corresponding to the BCC phase (rich in Fe) and γ-brass type structure (rich in Mn), respectively. The histogram shows the distribution of the powder particle with the milling time. The average powder particle decreases with the prolongs milling time.

Table 3
Elemental composition of 35 h milled powder samples by full area and point EDS analysis.

Sample condition	Scan area	Elemental composition (at%)				
		Fe	Mn	Ni	Al	Si
35 h milled powder	Full area	42.0	21.0	13.0	13.0	9.0
	Spot 1	54.0	22.0	14.0	6.0	4.0
	Spot 2	34.0	61.0	0.6	4.0	0.2

SPSed pellet of non-equiatomic HES, and the distribution of all the elements are uniform and the chemically homogeneous. The BSE image and the EPMA elemental mapping of the SPSed sample reveals the multi-phase nature of the non-equiatomic HES as shown in Fig. 8. The different contrast features are clearly visible in the BSE image i.e., white, light grey and evenly distrubuted dark grey particles. The point WDS analysis of these SPSed samples revealed that the white contrast was found to be rich in Al and Ni, which correspond to B2-type phase (spot 1). The light grey contrast were found to be rich in Fe, Mn and Ni, corresponding to BCC (ferritic) phase (spot 2). The values of the point WDS analysis of the elemental weight percentages are listed in Table 5.

Further, Fig. 8 represent the EPMA elemental maps of Fe, Mn, Ni, Al and Si. It can be discerned that the Fe, Mn and Ni was found to be almost uniformly distributed in the matrix owing to the exsistence of the major BCC phase. The high X-Ray intensity regions corresponding to the Al has slightly higher intensity of Fe and Ni, which may be attributed to the presence of B2-type phase. Similarly, the region enriched with Si shows the presence of Fe_5Si_3-type phase. It is very dificulty of distinguish the regions corresponding to the γ-brass type phases. This can be corroborated with the XRD results owing to the very less phase fraction of γ-brass type phases in SPSed samples. The EMPA elemental mapping

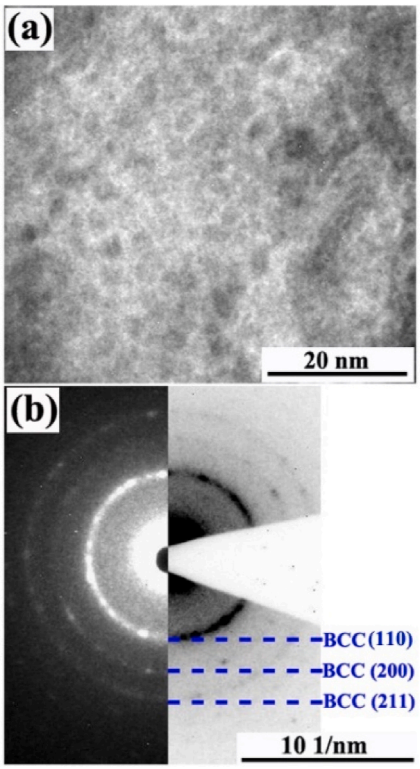


Fig. 3. (a) The TEM bright-field image of the 35 h milled sample; (b) its corresponding selected area diffraction pattern. Polycrystalline ring pattern corresponds to the BCC phase.

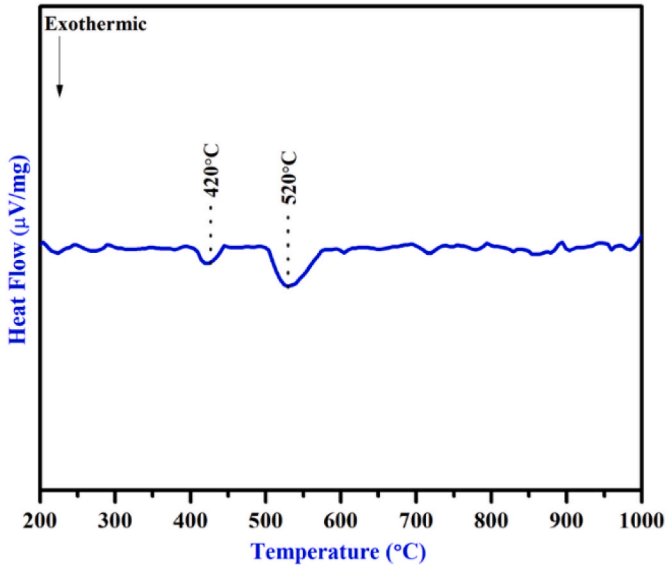


Fig. 4. DTA thermogram of the 35 h milled powder of the high entropy steel from 200 °C to 1000 °C.

helped in the understanding and presence of the multi-phase microstructure in these HES prepared by SPS (Fig. 8).

The load-displacement curve of the SPSed pellet of non-equiatomic high entropy steel is shown in Fig. 9(a). The nature of the p-h curve depicts the presence of limited plasticity in these sintered samples. The depth of indetation for the SPSed samples was found to be ~5.08 μm. Fig. 9(b) shows the optical image of indent on the surface of SPSed pellet. The optical micrograph shows no evidence of crack formation even at a load of 5000 mN. Fig. 9(c) display the compressive stress-strain plot of the SPSed sample. The evaluated values of the average microhardness and elastic modulus of the SPSed pellet are observed approximately 7.8 ± 0.3 GPa and 184 ± 5 GPa, respectively. Fig. 9(d) shows the radar diagram of the SPSed pellet revealing the physical and mechanical properties. The evaluated values of the average ultimate compressive stress and strain of the SPSed pellet are 2046 ± 160 MPa and 19%, respectively. Fig. 9(e) and (f) show the comparative curve between microhardness and compressive strength vs theoretical density, respectively of the present alloy and the reported HEAs, conventional stainless steel, dual-phase steel and low-density steel.

4. Discussion

4.1. Alloying behavior and microstructure

The phase formation during milling and after spark plasma sintering was correlated with the thermodynamic parameters. Zhang et al. [42] proposed the three parameters for the prediction of disordered solid solution formation in the HEAs i.e., high configurational entropy (ΔS_{conf}), mixing enthalpy (ΔH_{mix}) and the size mismatch (δ). The ΔS_{conf}, ΔH_{mix} and δ parameters were calculated based upon the Boltzmann hypothesis, Miedema model and Zhang's approach respectively [38,42]. The calculated values of thermodynamic parameter for the non-equiatomic high entropy steel and the criteria to form the solid solution are listed in Table 6. The values of the various parameter are well within the range of random solid solution formation criteria. Guo et al. [43] proposed some more parameters i.e., electronegativity (χ), valence electron concentration (VEC) and Ω for the prediction of solid solution formation in the multicomponent alloys. The VEC is used to predict the FCC/BCC or both the phase formation in the HEAs. The VEC is well within the range of dual-phase formation, and it forms the BCC type structure in the milled and SPSed sample of the present alloy. $Al_xCoCrCuFeNi$ (x = 0.45, 1, 2.5, 5 at. %) HEAs phase formation were discussed on the basis of mixing enthalpy, configurational entropy and size mismatch [44]. It was stated that the Al increased the B2 phase formation due to the high mixing enthalpy in the HEAs. It is also mentioned that the enthalpy of mixing in the HEAs system was preferred to form the ordering or clustering (formations of intermetallic compound) [45]. Shivam et. al [46], discussed the phase formation in the non-equiatomic $Fe_{40}Cr_{25}Ni_{15}Al_{15}Co_5$ medium entropy alloy in terms of binary mixing enthalphy and configurational entropy, which has formed the BCC and B2 structure. The B2 phase formation in the alloy was correlated with the binary enthalphy of mixing in Ni–Al which is the highest among the other pairs. In present alloy, the binary enthalpy between the Al–Fe, Al–Ni are highly negative. The values of the various possible pair of binary enthalphy are mention in the Table 1. To retain the solid solution in the HEAs at room temperature, the configurational entropy is not sufficient. The partially ordered B2-type structure in the alloy was formed due to the binary negative enthalpy. This parameters are well within the solid solution formation criteria. The criteria proposed for the prediction of the disordered solid solution in the HEAs are based upon the results of the reported HEAs. However, some of the reported HEAs suggested that the Zhang's criteria were neither necessary nor sufficient condition to predict the formation of disordered solid solution in the HEAs. This can only provide the basic idea to predict the phase formation criteria of disordered solid solution. Other than the thermodynamic parameters, the processing tenchique and atomic percentage of the indiviual elements also play an important role in the phase formation of the HEAs [45].

The diffraction intensity of the elemental peaks start diminishing as the milling time progresses, which indicates that the alloying is initiated. The dissolution of the individual elements during the milling is correlated with the physicochemical properties and the values are compiled in Table 7. The dissolution of the elements depends on the lowest melting point and the brittleness of the HEAs [47]. But, carbon, being interstitial element dissolved faster as compared to the other alloying

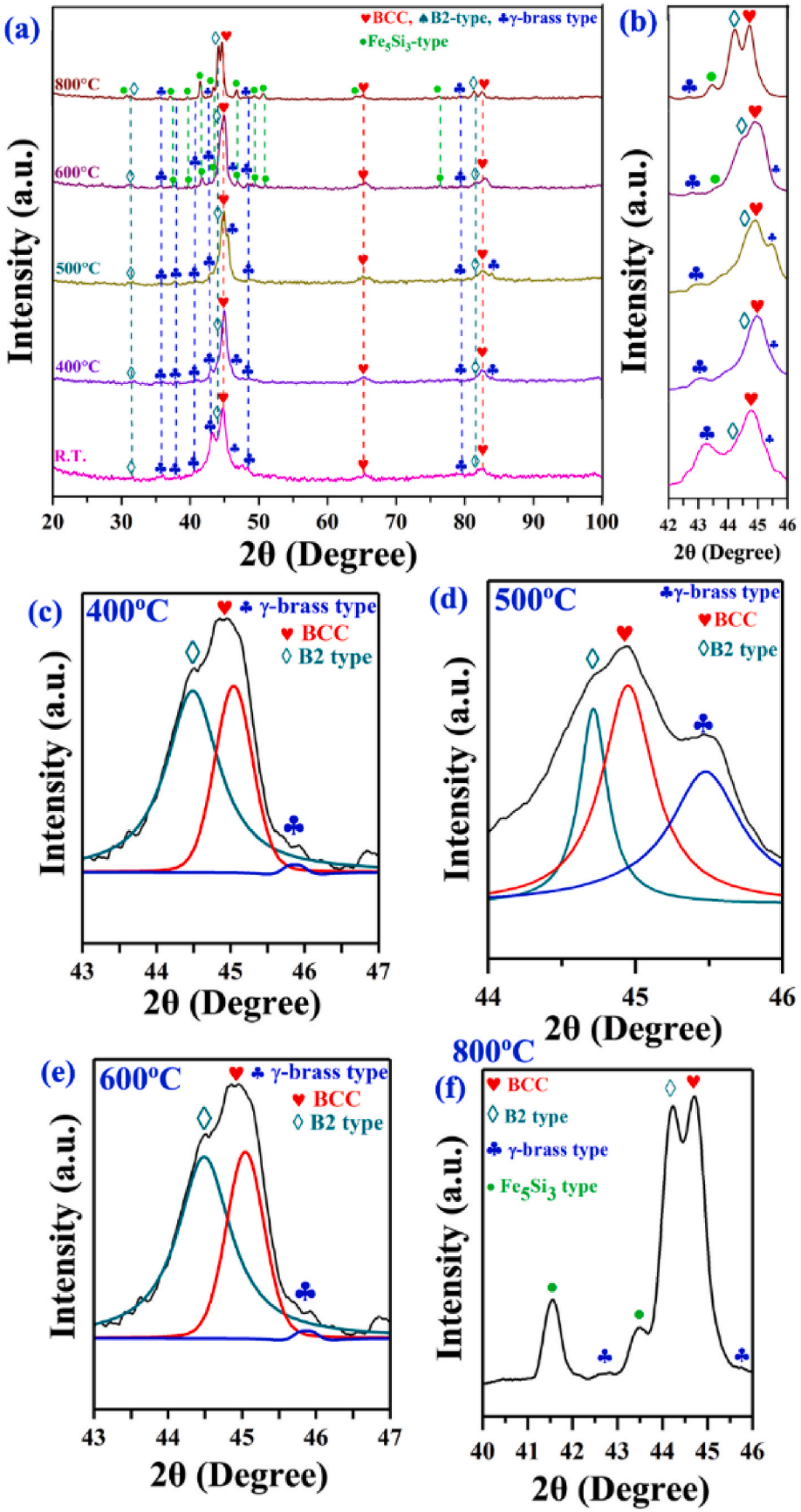


Fig. 5. (a) Ex-situ XRD diffraction pattern of the milled sample at R.T., 400 °C, 500 °C, 600 °C and 800 °C; (b) Enlarge view of diffraction peaks in 2θ range of 42°-46°; (c–e) Deconvoluted peaks for the 400 °C, 500 °C and 600 °C, respectively; (f) Enlarged view of the 800 °C, ranging from 40°- 46°. The formation of Fe_5Si_3-type intermetallic phase at 600 °C is correlated with exothermic event at 520 °C (Fig. 4).

Table 4
The phase fraction, lattice parameter and crystallite size of the annealed sample at 400 °C, 500 °C, 600 °C, 800 °C and SPSed samples of the non-equiatomic HES.

Condition	Phases	Phase Fraction	Lattice Parameter (nm)	Volume (Å^3)	Crystallite Size (nm)
400°C; wR = 2.088, GoF = 1.05, χ^2 = 1.10	BCC	57.00	0.286	23.39	21
	γ-brass type	21.30	0.872	663.05	12
	B2	21.70	0.289	24.12	18
500°C; wR = 2.279, GoF = 1.04, χ^2 = 1.07	BCC	47.70	0.286	23.39	23
	γ-brass type	36.30	0.872	663.05	16
	B2	16.00	0.289	24.13	20
600°C; wR = 0.252, GoF = 1.18, χ^2 = 1.40	BCC	43	0.286	23.39	26
	γ-brass type	3.00	0.872	663.05	19
	B2	38.50	0.289	24.13	21
	Fe_5Si_3 type	15.50	a = b = 0.667, c = 0.466	179.54	15
800°C; wR = 2.84, GoF = 1.29, χ^2 = 1.67	BCC	42.00	0.286	23.39	28
	γ-brass type	4.00	0.872	663.05	22
	B2	40.00	0.289	24.13	25
	Fe_5Si_3 type	14.00	a = b = 0.667, c = 0.468	180.31	20
SPSed; wR = 3.326, GoF = 1.33, χ^2 = 1.78	BCC	56.00	0.286	23.39	29
	B2	34.00	0.289	24.013	27
	γ-brass type	1.60	0.872	663.05	21
	Fe_5Si_3 type	8.40	a = b = 0.667, c = 0.468	180.31	23

elements present which might be due to its less atomic size and high diffusivity [11,35]. As the milling proceeds, Al dissolved followed by Mn and Ni and formed the solid solution. This alloying rate depends upon the melting point and self-diffusion coeffecient of the indiviual elements [47]. Si remains undissolved after 35 h of milling was due to the lowest self-diffusion coefficient and non-metallic behavior [47]. The lattice parameter of the major BCC phase, formed during the milling is close to the ferritic iron (α-Fe) (acting as host lattice). Shivam et al. [18] explained the concept of the host lattice in HEAs system in terms of melting point and the bond strength of the elements present in the alloys.

4.2. Thermal stability

The phase evolution of the non-equiatomic HES at higher temperature are observed at 25 °C such as BCC + γ-brass type + B2 + Si (undissolved). The alloy is thermally stable up to ~500 °C. At 420 °C, the phases observed to be stable are BCC + γ-brass type + B2-type. But the phase fraction of the γ-brass type structure increased and some peaks were evolved. This type of transformation was also seen in the non-equiatomic $Fe_{40}Cr_{25}Ni_{15}Al_{15}Co_5$ medium entropy alloy [46] and quasicrystals [29,48,49]. At 520 °C, the phases are found to be BCC + γ-brass type + B2 and Fe_5Si_3 type silicide. The silicide formation was also seen in the previously studied HEAs i.e., non-equiatomic FeMnNiTiAlSi–C [35], FeMnNiCrAlSi–C [11], AlCoCrFeNiSi [50] and AlCrFeMgNi [51] HEAs fabricated using the powder metallurgy route. The phase transformations occurred at higher temperature were due to the formation of metastable solid solution phase during milling. $Fe_{36}Mn_{21}Cr_{18}Ni_{15}Al_{10}$ high entropy alloy, the phases formed was BCC and B2. The alloy showed thermal stability of B2 phase at different heat treatment of the as-cast samples. The good thermal stability of the B2 phase was due to the strong bonding between the Ni–Al atoms and

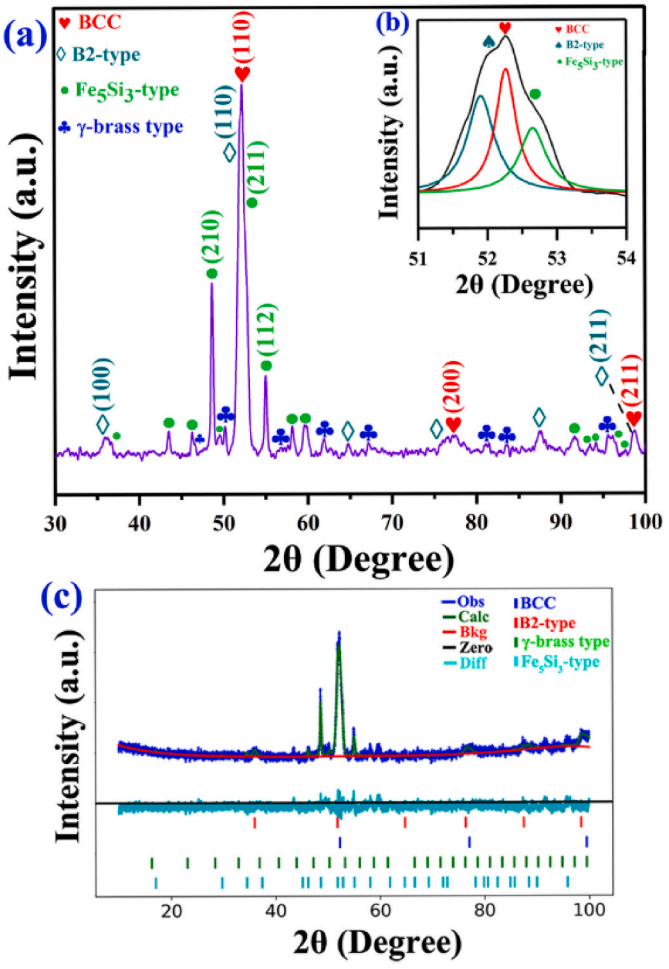


Fig. 6. (a) XRD pattern of the SPSed sample; (b) Deconvulation of the SPSed sample ranging from 51°- 54°; (c) Rietveld refinement plot of the SPSed sample. The ferritic (BCC) type high entropy steel with B2, γ-brass type structure and Fe_5Si_3 type intermetallics can be depicted.

sluggish diffusion effect in the high entropy alloys [46,52]. In the present alloy, the retention of the BCC and B2 phase after the various annealing condition and sintering was due to the strong bonding between the pairs of Fe–Al and Ni–Al, and sluggish diffusion effect. The formation of the silicide was due to the undissolved Si remaining after milling, and tendency of Fe and Mn to react with Si. The binary enthalpy of mixing between Fe–Si and Mn–Si is highly negative.

Fig. 10 shows the property diagram of the non-equiatomic HES, generated by the CALPHAD modelling (using Thermo-Calc software). The sequence of the phases in the predicted phase diagram generated using the Thermo-calc are as follows:

- The liquid (100%) formed at around 1200 °C which contains all the elements of the alloy system i.e., Fe, Mn, Ni, Al, Si and C.
- Around 1100 °C, from the liquid, the two phases formed i.e., BCC_A2#1 phase (which contains Fe, Mn, Ni, Al, Si and C) and B2 (contains Ni and Al).
- Around 800 °C, number of phases, formed remained same and after this the BCC_A2#2 phase formed (contains Mn and C).
- In between 800 °C and 500 °C, there was no phase change after this but the M_3Si intermetallic compound formed which was enriched in Fe, Mn and Si.
- Around 300 °C, the BCC_A2#1, B2, BCC_A2#2 and M_3Si were retained and the new phase was evolved- BCC_A2#3, which was

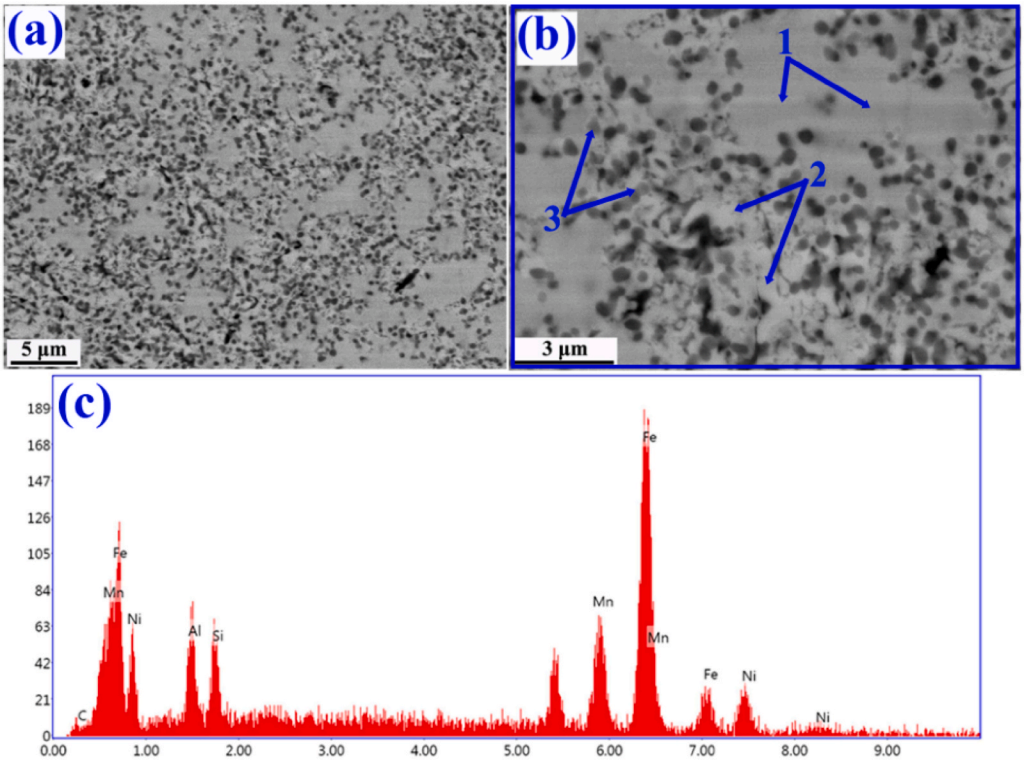


Fig. 7. (a) and (b) SEM image of the SPSed sample of the non-equiatomic high entropy steel at different magnifications; (c) EDS spectrum of SPSed HES by Full area EDS analysis of the marked area (in (b)). The different contrast i.e., white (2), light grey (1), dark grey (3) contrast are clearly visible in SEM images. The shape of black and dark grey are onion type.

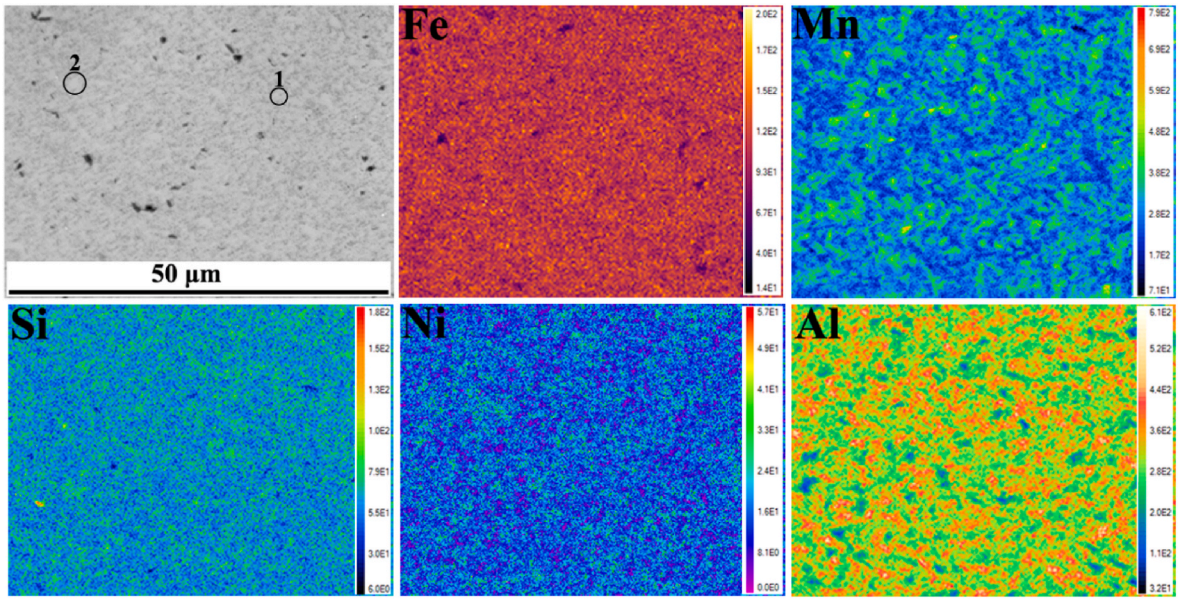


Fig. 8. BSE image and EPMA elemental mapping of SPSed sample.

Table 5
Elemental composition of SPSed sample of the non-equiatomic HES by point WDS analysis.

Sample condition	Scan area	Elemental composition (wt.%)				
		Fe	Mn	Ni	Al	Si
SPSed sample	Spot 1	32.46	10.00	32.09	21.82	2.63
	Spot 2	44.53	21.60	20.66	7.33	5.88

enriched in Fe, Mn and Si. After this, the phases remains unaltered up to room tempearture.

Table 8 listed the phases form from the 25 °C–1200 °C, and its mole fraction, which was predicted through the single-point equilibrium calculation of the alloy using the Thermo-Calc (TCFE8 database). The BCC, partially ordered B2 and γ-brass type structure were formed during the milling and it was correlated with the CALPHAD predicted phases.

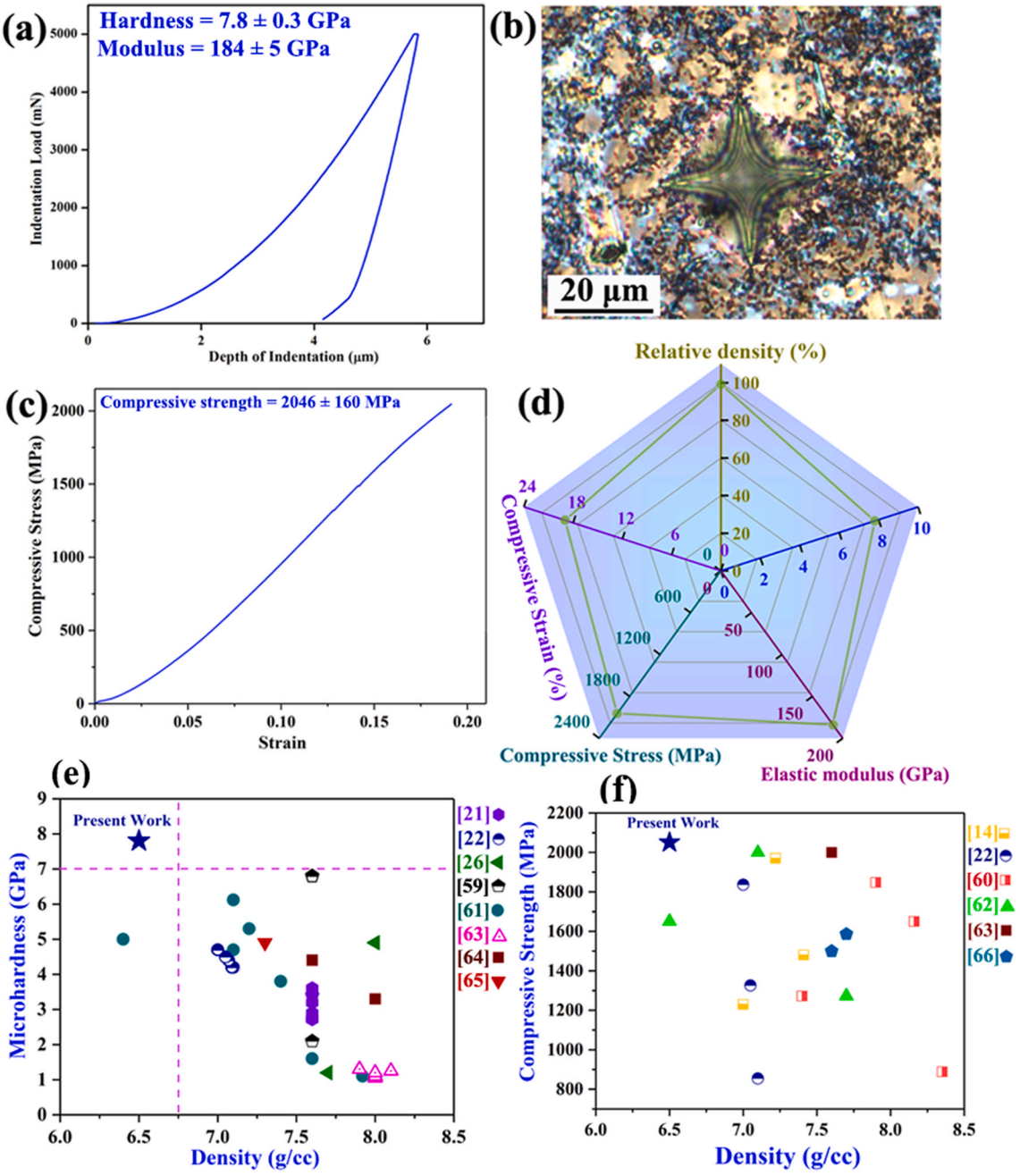


Fig. 9. (a) Depth of penetration vs Indentation load (p-h plot); (b) Optical micrograph of the indent on SPSed HES sample; (c) Compressive engineering stress and strain curve; (d) Radar diagram showing comparision of different mechanical properties; (e and f) Comparition (microhardness vs density, and ultimate compressive stress vs density) of the present work with the conventional steel and other HEAs (details mentioned in supplementary Table S1). This illustrate the physical and mechanical properties of the SPSed pellet.

Table 6
Theortical criteria of the thermodynamic parameter for the regular solid solution formation in the HEAs and the thermodynamic parameters of the non-equiatomic FeMnNiAlSiC HES.

Thermodynamic parameter	ΔH_{mix} (kJ/mol)	ΔS_{conf} (J/K-mol)	δ (%)	Ω (Dimensionless)	VEC (Dimensionless)	References
Regular solid solution	$-22 \leq \Delta H_{mix} \leq 7$	$11 \leq \Delta S_{conf} \leq 19.5$	$\delta \leq 6.6$	$\Omega \geq 1.1$	VEC ≥8.0 FCC	[42,43]
Present work	−8.4	12.4	7.6	2.4	VEC <6.87 BCC 6.87 < VEC <8.0 BCC + FCC 6.9	This work

Table 7
Physiochemical properties of constituent elements in non-equiatomic FeMnNiAlSiC HES.

	Fe	Mn	Ni	Al	Si	C
Atomic radius (Å)	1.27	1.40	1.24	1.40	1.11	0.91
Density (g/cc)	7.87	7.4	8.91	2.70	2.3	2.1
Melting point (°C)	1536	1244	1455	660	1414	3550
Self-diffusion coefficient (m^2/s)	10^{-31}	10^{-36}	10^{-36}	10^{-19}	10^{-62}	–

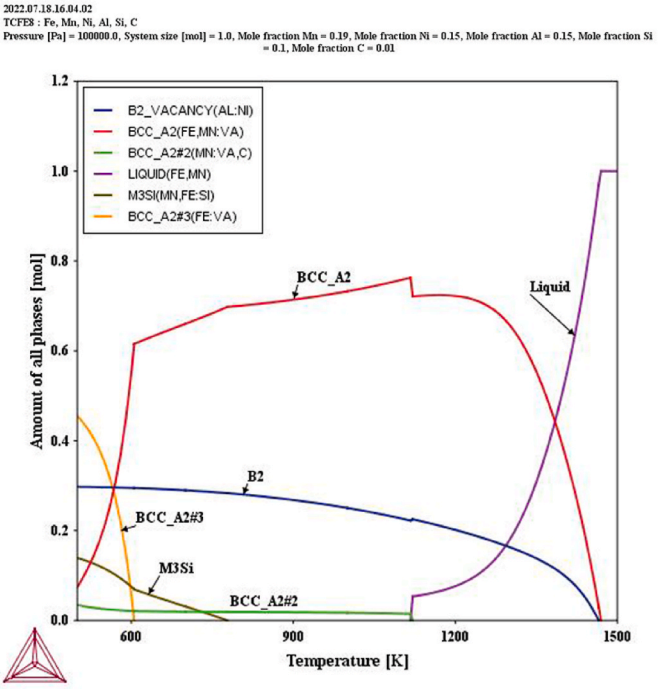


Fig. 10. Property diagram of high entropy steel through CALPHAD modelling.

Table 8
Single-point equilibrium of the high entropy steel at different temperature by CALPHAD modelling (Thermo-Calc software).

Condition	Predicted Phases	Mol
1200°C (1473 K)	Liquid (contains Fe, Mn, Ni, Al, Si, C)	1.0
1100°C (1373 K)	Liquid	0.40
	BCC_A2#1 (contains Fe, Mn, Ni, Al, Si, C)	0.48
	B2 (contains Ni, Al)	0.12
800°C (1073 K)	BCC_A2#1	0.75
	B2	0.23
	BCC_A2#2 (contains Mn, C)	0.02
500°C (773 K)	BCC_A2#1	0.69
	B2	0.28
	BCC_A2#2	0.02
	M_3Si (contains Fe, Mn, Si)	0.01
300°C (573 K)	BCC_A2#1	0.26
	B2	0.29
	BCC_A2#2	0.02
	M_3Si	0.11
	BCC_A2#3 (contains Fe, Mn, Si)	0.32

The phases formed during the annealing and SPSed were also correlated with the CALPHAD (Thermo-Calc) modelling predicted phases. Some difference was there between the experimental results and CALPHAD predicted phases. This may be due to the fact that the CALPHAD database TCFE8 was used instead of the TCHEA3. Therefore, the differences in the predicted and the experimental results was due to the limitations of the CALPHAD database used here.

4.3. Mechaincal properties and strengthening mechanisms

The measured yield stress of the SPSed sample of the non-equiatomic HES is found to be 2046 ± 160 MPa and it is the combination of the frictional stress and the various strengthening mechanisms including grain-boundary, precipitates, dislocation and solid solution strengthening. The total yield stress can be expressed as:

$$\sigma_y = \sigma_o + \Delta\sigma_{gb} + \Delta\sigma_p + \Delta\sigma_d + \Delta\sigma_{ss} \quad (5)$$

where, σ_o is frictional stress, $\Delta\sigma_{gb}$ is grain boundary strengthening, $\Delta\sigma_p$ is precipitate strengthening, $\Delta\sigma_d$ is dislocation strengthening and $\Delta\sigma_{ss}$ is solid solution strengthening. The frictional stress is taken as 120 MPa for BCC and B2, respectively [53]. The grain boundary strengthening was calculated using the Hall-Petch relation [35] as,

$$\Delta\sigma_{gb} = K_o d^{-½} \quad (6)$$

where, K_o is the constant related to the material and value is taken as 574 MPa/μm$^{1/2}$ for BCC and partially ordered B2-type respectively [54] and *d* represents the grain size. The value of the grain size for BCC and B2-type was found to be 0.28 μm and 0.22 μm, respectively. The value for grain boundary strengthening was evaluated to be 1085 MPa and 1224 MPa for BCC and B2-type, respectively.

The effect of the silicide and the B2-type precipitate on the strength of the high entropy steel was calculated using the equation [35] as:

$$\Delta\sigma_p = \frac{0.4MGb}{\pi\lambda}\frac{\ln\left(\frac{2\bar{r}}{b}\right)}{\sqrt{1-\upsilon}} \quad (7)$$

where, M is the Taylor factor (2.73) [53], G is the shear modulus, calculated using the rule of mixture and its value was found to be 75.34 GPa, υ is the Poisson's ratio (0.3) [35], b is the Burgers vector ($\frac{\sqrt{3}}{2}$ *a*), a is the lattice parameter for BCC, λ is the spacing between the precipitates, calculated using the equation as, $\lambda = 2\bar{r}\left(\sqrt{\frac{\pi}{4f}} - 1\right)$, f is the volume fraction of the precipitates (8.4% for Fe_5Si_3-type and 34% for B2-type), $\bar{r}$ is the average radius (circular cross section) of the spherical precipitates, calculated using the equation as, $\bar{r} = \sqrt{\frac{2}{3}}r$ and r is the average radius of the precipitates (102 nm for Fe_5Si_3-type and 220 nm for B2-type). The value of the dispersion strengthening for BCC phase was found to be 54 MPa and 111 MPa for silicide and B2-type, respectively and for B2-type was found to be 55 MPa for silicides.

The contribution of the dislocation strengthening was calculated using the Bailey-Hirsch relationship [35] as:

$$\Delta\sigma_d = M\alpha Gb\sqrt{\rho_{dd}} \quad (8)$$

where, α is the constant and taken as 0.2 for BCC and B2-type [35], respectively, b is the Burgers vector and ρ_{dd} is the dislocation density, calculated using equation (2) and the value was taken as 4.46 × 10^{15} m^{-2} and 3.36 × 10^{15} m^{-2}, for BCC and B2-type, respectively. The strength value was found to be 687 MPa and 590 MPa for BCC and B2-type respectively. The solid solution strengthening was calculated using the Fleischer model [35] as:

$$\Delta\sigma_{ss} = \frac{MG\varepsilon_f}{700}c^{1/2} \quad (9)$$

where, c is the atomic percentage of the solute element and ε_f is the relation parameter and given as, $\varepsilon_f = \left|\frac{\varepsilon_G}{1+0.5|\varepsilon_G|} - 3\varepsilon_a\right|^{3/2}$, ε_G is the modulus misfit ($\frac{1}{G}\frac{dG}{dc}$) and ε_a is the atomic size misfit ($\frac{1}{a}\frac{da}{dc}$). Al was taken as solute element for both, the BCC and B2-type phases as Al has highest

atomic radius as compared to the other elements and the atomic percent was taken as 13.45 and 36.55. The strength value was found to be 106 MPa and 177 MPa for BCC and B2-type, respectively.

The strengthening mechanism based calculation was done on the assumption that the phase fraction of the BCC and B2-type was 100%. The phase fraction of BCC and B2-type in the SPSed sample of the high entropy steel was calculated as 56.0% and 34.0%, respectively. Therefore, the phase fraction of the BCC and B2-type for the calculation of the strengthening mechanism was taken as 62.0% and 38.0%, respectively. The contribution of the various strengthening mechanism is mentioned in Table 9. The grain boundary and dislocation strengthening are dominant strengthening mechanisms. The individual contribution of different strengthening mechanisms are as follows: grain boundary (52.55%), dislocation (30.00%), precipitation (5.70%) and solid solution strengthening (6.14%). The calculated theoretical strength and the measured strength of the alloy are in close agreement.

It was observed that the addition of the Si promotes the formation of BCC and silicides phases with Cr, Fe, Ni and Ti in the HEAs. The mechanical properties i.e., hardness and compressive strength was increased with the addition of the silicon in the reported HEAs [55]. $Al_{0.3}CoCrFeNiSi_x$ showed the phase transition from FCC to BCC/ordered B2 phase as x increases (0 to 1). The tensile strength and hardness of the HEAs increased without any decrease in the tensile ductility by 63% and 143 Hv to 863 Hv as silicon content increased (x = 0–1 at. %) [56]. Wang et al. [57] reported that the formation of M_5Si_3 type silicide in the HEAs enhanced the strength and plasticity. The strength and plasticity both increased with the small content of Si. However, as Si increases, the strength also increases but plasticity decreases in the HEAs. In $Al_{0.2}Co_{1.5}CrFeNi_{1.5}Ti_{0.5}$ HEAs, the silicide formation plays a crucial role to strengthen the alloy and it increases the hardness and compressive strength of the alloys [58]. In the non-equiatomic FeMnNiCrAlSi–C high entropy steel, which has formed the FCC and BCC dual-phase structure with the carbide and silicide intermetallic compound, showed the hardness value of 5.86 GPa [11]. In contrast to previous work on the high entropy steel, the present non-equiatomic high entropy steel has showed the high hardness value of 7.8 GPa. In the present alloy, the in-situ formation of the silicide has played a crucial role to strengthen the alloy i.e., microhardness and compressive strength with good amount of ductility.

A comprehensive comparison of the present alloy with the high entropy alloys is shown in Fig. 9(e and f). In addition, the comparison with the conventional stainless steel, dual phase steel and low-density steel is also shown. The ferritic dual phase steel has the microhardness in the range of 2.0–2.4 GPa [59]. The microhardness of the present alloy (7.8 GPa) is much higher than the dual-phase steel. The Fig. 9 (e and f) shows the comparison of different alloys with respect to theoretical density and microhardness with the reported HEAs [21,22,26,59,60–66], which are also mentioned in the Supplementary Table S1. The quinary low-density high entropy steel with the formation of BCC and B2 with silicide phase (intermetallic compound) showed the excellent mechanical properties i.e., compressive strength and microhardness as compared to the HEAs and conventionally steel. The present investigation envisaged to fabricate a low-density high entropy steel possessing good microstructural stability as well as excellent mechanical properties having synergy with ductility. In spite of having excellent microstructural stability and mechanical properties, these SPSed HES were found to have a limited ductility of ~19%. Further, efforts towards tailoring the composition for designing and developing a low-density HES having excellent room temperature and high temperature properties may be a way forward for its potential structural applications.

5. Conclusion

The phase evolution of the $Fe_{40}Mn_{19}Ni_{15}Al_{15}Si_{10}C_1$ high entropy steel after high energy ball milling, and their thermal, structural and microstructural stability after spark plasma sintering were evaluated.

Table 9
The value of calculated theoretical strength and the measured strength of the non-equiatomic high entropy steel.

Strengthening Mechanism	Phases	
	BCC	B2-type
Frictional stress (σ_o)	120 MPa	120 MPa
Grain Boundary ($\Delta\sigma_{gb}$)	1085 MPa	1224 MPa
Dislocation ($\Delta\sigma_d$)	687 MPa	590 MPa
Precipitation ($\Delta\sigma_p$)	54 MPa (silicides) + 111 MPa (B2-type)	55 MPa
Solid solution ($\Delta\sigma_{ss}$)	106 MPa	177 MPa
Total contribution		**2164 MPa**
Experimental strength		**2046 MPa**

Further, experiments were done to determine the mechanical properties i.e., microhardness, elastic modulus and compressive strength of the SPSed sample. The following conclusions can be drawn from the present study:

1. After 35 h of mechanical milling the phases formed were BCC, partially ordered B2-type and γ-brass type structure. The metastable alloy showed thermal stability up to 500 °C. Afterwards, the Fe_5Si_3-type intermetallic phases were formed at 520 °C. The formation of silicide at higher temperature could be due to higher binary negative enthlphy between the Fe–Si and atomic size mismatch, unfavourable for solid solution.
2. The BCC and B2-type phases retained after spark plasma sintering along with the formation of Fe_5Si_3-type silicide phase. The nanocrystalline structure was observed even after annealing treatment and spark plasma sintering of the powder samples.
3. The dual phase structure i.e., BCC and B2 showed good strength i.e., microhardness (7.8 ± 0.3 GPa) and ultimate compressive stress (2046 ± 160 MPa) with an appreciable amount of compressive ductility (19%). The calculated values of strengthening based on materials modelling was close to the experimental values. The grain boundary and dislocation strengthening are found to be the dominant mechanisms. The phase prediction through thermodynamic parameters and Thermo-Calc were correlated with the experimental results.

CRediT authorship contribution statement

Harsh Jain: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Validation, Writing – original draft. **Yagnesh Shadangi:** Methodology, Investigation, Validation, Formal analysis, Writing – review & editing. **Dibyendu Chakravarty:** Resources. **Kausik Chattopadhyay:** Resources, Methodology. **Ashutosh Kumar Dubey:** Supervision. **N.K. Mukhopadhyay:** Supervision, Conceptualization, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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